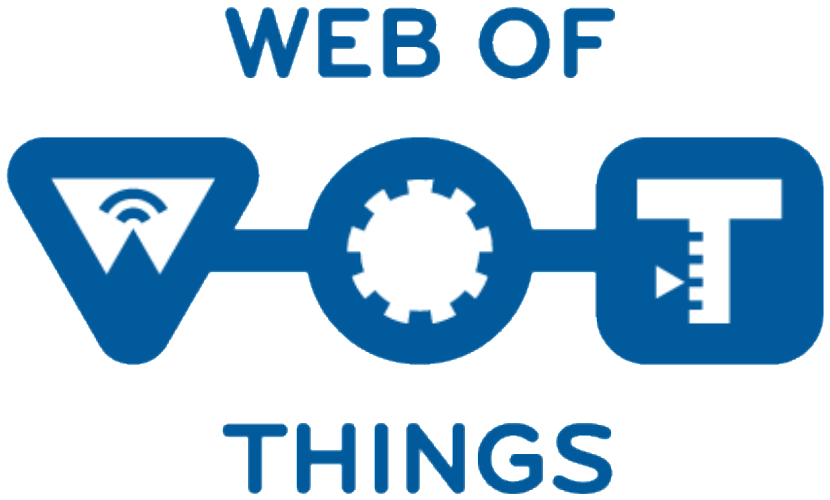


Thing Description Sessions

Ege Korkan, Siemens

29 October 2026

W3C TPAC 2026

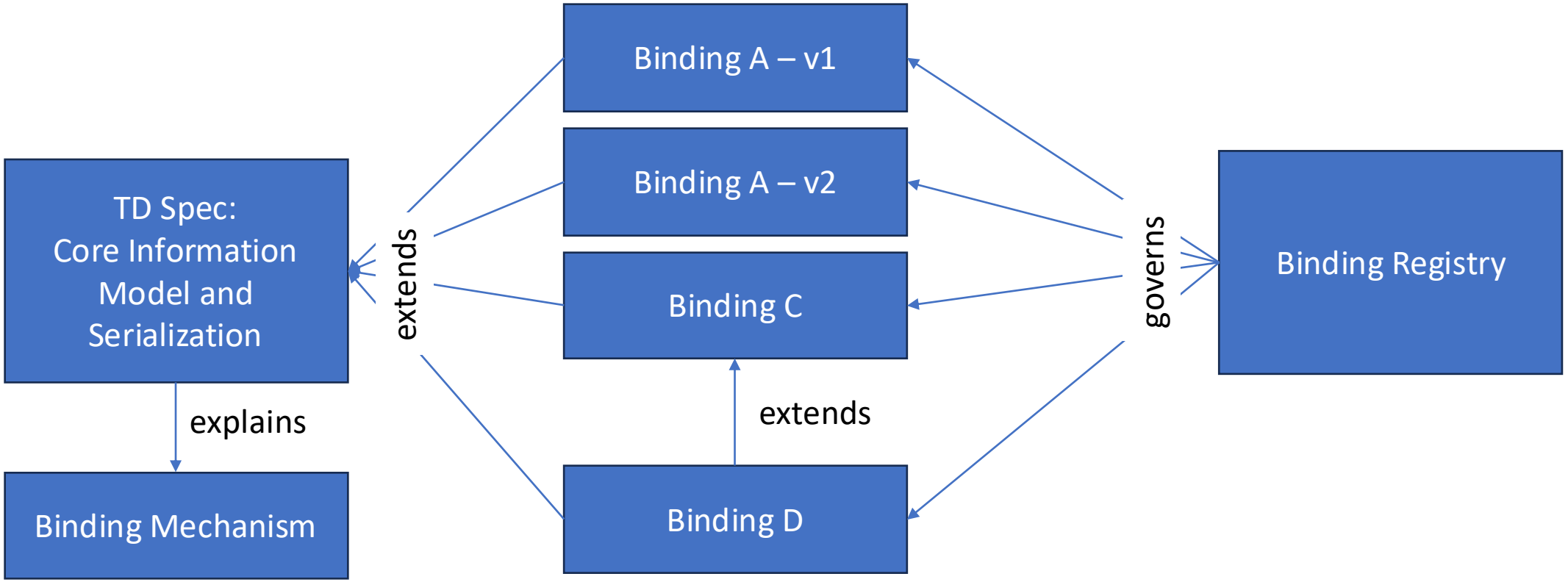
Agenda

- Today
  - Summaries (20 mins), Ege:
    - TD Work (common definitions, data mapping, underlying stuff, toolchain)
    - New bindings and changes
    - Binding Registry
  - Specification Toolchain (20 mins), Mahda
- Tomorrow
  - TD roadmap / Prioritization / Who is responsible of specific tasks (20 mins), Ege
  - Data Mapping User Stories to Specification Feature (15 mins), Christian
  - Subprotocols, URI Schemes, and how to identify and manage bindings (30 mins), Ege

Summaries

TD, Bindings, Registry

Landscape


TD Spec:
Core Information
Model and
Serialization
explains
Binding Mechanism
extends
Binding A – v1
Binding A – v2
Binding C
Binding D
extends
governs
Binding Registry

Thing Description

TD

- Getting Organized
- Common Definitions
- Data Mapping
- Media Streaming
- Core Design Challenges

Getting Organized

- Work Item Planning
  - https://github.com/w3c/wot-thing-description/blob/main/planning/README.md contains the summarized work items and analyses
- Architecture Specification Assertions
  - We worked on moving normative assertions in the Architecture specification to the TD
  - https://github.com/w3c/wot-thing-description/issues/2126 and was tracked by https://github.com/w3c/wot-thing-description/blob/main/planning/work-items/other-documents-sync/arch.md

Common Definitions

- First big refactoring and almost a feature of TD 2.0
- Waiting for toolchain refactoring to move to Editor's Draft
- https://github.com/w3c/wot-thing-description/tree/main/proposals/common-definitions
- Plugfest TDs with common definitions: https://github.com/w3c/wot-testing/pull/641
- Upcoming examples are from https://github.com/w3c/wot-thing-description/blob/main/proposals/common-definitions/tooling/tds.js

Common Definitions: Simple Case

```json
{
  "@context": "https://www.w3.org/ns/wot-next/td",
  "title": "valid-test-cbor-default",
  "formDefaults": {
    "contentType": "application/cbor",
    "base": "coap://[2001:DB8::1]/mything",
    "security": {
      "scheme": "nosec"
    }
  },
  "properties": {
    "prop1": {
      "type": "string",
      "forms": [
        {
          "href": "props/prop1"
        }
      ]
    },
    "prop2": {
      "type": "string",
      "forms": [
        {
          "href": "props/prop2"
        }
      ]
```

```json
{
  "@context": "https://www.w3.org/ns/wot-next/td",
  "title": "valid-test-modbus-params",
  "formDefaults": {
    "base": "modbus+tcp://192.168.178.32:502/1/",
    "modv:timeout": 1000,
    "modv:pollingInterval": 5000,
    "modv:zeroBasedAddressing": true,
    "modv:mostSignificantByte": true,
    "modv:mostSignificantWord": true,
    "contentType": "application/octet-stream",
    "security": {
      "scheme": "nosec"
    }
  },
  "properties": {
    "prop1": {
      "type": "boolean",
      "forms": [
        {
          "href": "1",
          "modv:entity": "Coil"
        }
      ]
    },
    "prop2": {
      "type": "string",
      "forms": [
        {
          "href": "2",
          "modv:entity": "HoldingRegister"
        }
      ]
```

Common Definitions: Multiple Form Defs.

```
{
  "@context": "https://www.w3.org/ns/wot-next/td",
  "title": "valid-test-multi-ip",
  "formDefinitions": {
    "ipv4": {
      "base": "https://192.168.1.10:8080",
      "contentType": "application/json",
      "security": {
        "scheme": "nosec"
      }
    },
    "ipv6": {
      "base": "https://[2001:db8:85a3::8a2e:370:7334]:8080",
      "contentType": "application/json",
      "security": {
        "scheme": "nosec"
      }
    }
  },
  "formDefaults": ["ipv4", "ipv6"],
  "properties": {
    "prop1": {
      "type": "string",
      "forms": [
        {
          "href": "props/prop1"
        }
      ]
    },
    "prop2": {
      "type": "string",
      "forms": [
        {
          "href": "props/prop2"
        }
```

Data Mapping

- Christian will go deeper into it
- Developed at https://github.com/w3c/wot-thing-description/blob/main/planning/work-items/analysis/data-mapping/analysis-data-mapping-summary.md

Media Streaming

- How can we handle media streams that are not simple fetching of media. Early analysis that will be extended from Plugfest results.
- Driven by Toumura-san
- https://github.com/w3c/wot-thing-description/blob/main/planning/work-items/analysis/analysis-media-streaming.md

Core Design Challenges

- Discussion about the fundamentals of the TD. Difficult to summarize each point ☺
  - Subprotocols, URI Schemes, and how to identify and manage bindings: Deeper discussion tomorrow
  - readOnly, writeOnly and how these influence input and outputs
    - Failed PR: https://github.com/w3c/wot-thing-description/pull/2206
    - Next discussion: https://github.com/w3c/wot-thing-description/issues/2215 and https://github.com/w3c/wot-thing-description/blob/main/planning/work-items/analysis/readOnly-writeOnly.md

Bindings

Bindings: Changes

Not much change to existing bindings

- HTTP Binding: Not anymore part of TD
  - https://github.com/w3c/wot-thing-description/pull/2123
- Modbus: Zero based addressing
  - https://github.com/w3c/wot-binding-templates/pull/479

Bindings: New Bindings

- LoRaWAN: https://w3c.github.io/wot-binding-templates/bindings/protocols/lorawan/index.html
- EtherNet/IP: https://w3c.github.io/wot-binding-templates/bindings/protocols/ethernetip/index.html
- S7Comm: https://w3c.github.io/wot-binding-templates/bindings/protocols/s7comm/index.html
- Other SDOs:
  - OPC UA: https://reference.opcfoundation.org/specs/OPC-10101
  - CAN: https://github.com/w3c/wot-binding-templates/issues/465
  - SNMP: https://github.com/w3c/wot-binding-templates/issues/464
  - IO Link: Link to be provided but it is in IDTA

All we have

- Binding Publications by WoT WG:
  - HTTP Binding
  - CoAP Binding
  - MQTT Binding
  - Modbus Binding
  - BACnet Binding
  - PROFINET Binding Template
  - S7Comm Binding
  - EtherNet/IP Binding
  - S7Comm Binding
  - LoRaWAN Binding
  - XML Binding
- Binding Publications by other organizations:
  - OPCUA Binding

https://github.com/w3c/wot-binding-templates/#publications

Binding Registry

Registry

- It is just a table!
- https://w3c.github.io/wot-binding-registry/
- However, we define how that table is governed. That makes things a bit more difficult and “administrative”
- Some support would be appreciated here

What goes into the table

Table 1 Entry format information

Characteristic	Type	Uniqueness	Description
Name of the Binding	`string`	Required	Human-readable name for the binding. Examples: `HTTP Binding`, `CoAP Binding`
Version	`string`	Required within one Binding's lifecycle	A string that identifies a specific version of this binding. Example: `1.0-2026-01-15`. The version string *SHOULD* contain a UTC-based date in ISO 8601 format in the form of `YYYY-MM-DD`.
Status	enumeration of [`Initial`, `Current`, `Superseded`, `Obsolete`]	Not required	Indicates the lifecycle status of the entry and changes via transitions defined in Lifecycle section.
Binding Document Link	`anyURI`	Required	Stable link whose content cannot change (e.g., a date, version number, etc.), which can be managed by another entity than the custodian. Examples: `https://www.w3.org/TR/wot/binding-templates/http-20240726/index.html`
Target Specification of the Binding	`string` followed by an `anyURI`	Required	The name and link of the specification that the binding is designed to target. Example: `MQTT 3.1.1 (https://docs.oasis-open.org/mqtt/mqtt/v3.1.1/os/mqtt-v3.1.1-os.html)`
Summary Document Link	`anyURI`	Required	Link to the summary document of the Binding.
Vocabulary Term Prefix	`string`	Required	Prefix used for vocabulary terms such as `htv` for HTTP and `modv` for Modbus
Binding Identification in TD	any type	Required	URI Scheme or other TD terms for a Consumer to identify the binding. Examples: `"subprotocol":"sse"`, `"href":"https://example.com"`, `"contentType":"application/json"`
Supported TD Versions	Array of `string`	Not required	The versions of TD Recommendations that the binding supports. The allowed values are reflected in the submission templates. Example: `["1.0", "1.1"]`.

Some rules in the registry

§ **4.2.1 Administrative Requirements**

- **Stable Links:** The link to the binding document and its supporting documents *MUST* guarantee the retrieval of the same document over time. This allows the one managing the hosting of the binding document to add redirection rules in case the hosting location needs to be changed, while still guaranteeing the stability of the link and the immutability of the content.
- **Immutable Content:** The submitter *MUST* guarantee that the content of the specific version of the binding document will not change over time.
- **Copyright:** The WoT binding document *MAY* follow another copyright than the W3C copyright. The submitter is free to choose based on the process they or their organization follows.
- **Open to Read:** The binding document linked in the registry entry *SHOULD* be open to read, use, and implement, but that is not required for the document to be added to the registry.
- **Reviewer Access:** Reviewers *MUST* have access to the binding document and to the protocol or media type specification (what the binding specifies).

Toolchain

Presentation by Mahda – Switch to her screen

Roadmap

This should be a very open discussion and understanding the contribution plans of everyone. Who can do what, who **wants** to what?

What needs to be done

Bold: Someone takes care of it
Italic: Shared work or cannot be assigned to one
Underline: Need support
No style: No one takes care of it

Thing Description:

- **Toolchain – Mahda, Abror**
- *Core design changes*
- *Simple improvements*
- **Common Definitions - Ege**
- **Data Mapping - Christian**
- **Media Streaming – Toumura-san**
- Historical Data – Rob Warren
- Manageable Affordances
- Signing and Canonicalization
- Splitting Security into Bindings – Kaz and Matsumura

Generic Tasks:

- **Overall meeting organization - Ege**
- **Publication of Specs and of Resources - Ege**
- **Triaging Issues and Work - Ege**

Bindings-related:

- *Binding Authoring*
- Binding Review (via Registry or not)
- Registry document writing - Ege
- Publication of Bindings - Sebastian

Best practices & Primer:

TBD but relates to the TD Task Force

Data Mapping

Presentation by Christian – Switch to his screen

Subprotocols, URI Schemes, and how to identify and manage bindings